

ASSIGNMENT – 2

Basic Mathematics

B.Sc (Hons.) Agriculture / B.Tech Biotechnology

Beginner Level

Q1. Define:

- (a) Natural Numbers
- (b) Whole Numbers
- (c) Integers

Q2. Simplify:

$$\sqrt{72} - \sqrt{32}$$

Q3. Solve:

$$|3x + 2| = 11$$

Intermediate Level

Q4. Prove that:

$$\sqrt{5}$$

is irrational.

Q5. Factorize:

$$x^2 - 9$$

Q6. Solve:

$$x^2 + 7x + 10 = 0$$

Advanced Level

Q7. Find all roots:

$$x^4 - 13x^2 + 36 = 0$$

Q8. Prove:

$$\sqrt{2} + \sqrt{7}$$

is irrational.

Beginner Level

Q1. Define:

- (a) Finite Set
- (b) Infinite Set
- (c) Empty Set

Q2. Let:

$$A = \{2, 4, 6, 8\}$$

$$B = \{1, 2, 3, 4\}$$

Find:

- (a) $A \cup B$
- (b) $A \cap B$
- (c) $B - A$

Intermediate Level

Q3. Draw Venn diagrams for:

- (a) $A - B$
- (b) $B - A$

Q4. Verify:

$$(A \cup B)' = A' \cap B'$$

Advanced Level

Q5. In a survey:

- 80 students like Mathematics
- 50 like Physics
- 30 like both

Find:

- (a) Students liking only Physics
- (b) Students liking at least one subject

Beginner Level

Q1. Define relation and function.

Q2. Find domain:

$$f(x) = \frac{1}{x-3}$$

Q3. Evaluate:

$$f(x) = 3x^2 - 5x + 1$$

Find:

$$f(1)$$

Intermediate Level

Q4. Determine whether:

$$f(x) = x^2$$

is one-one.

Q5. Sketch graph:

$$y = |x + 2|$$

Advanced Level

Q6. Find inverse function:

$$f(x) = \frac{x+1}{2}$$

Q7. Check whether:

$$f(x) = x^3 + x$$

is onto.

Beginner Level

Q1. Define:

(a) Limit

(b) Continuity

Q2. Evaluate:

$$\lim_{x \rightarrow 1} (x^2 + 5x)$$

Q3. Evaluate:

$$\lim_{x \rightarrow 0} \frac{\tan x}{x}$$

Intermediate Level

Q4. Evaluate:

$$\lim_{x \rightarrow 3} \frac{x^2 - 9}{x - 3}$$

Q5. Check continuity:

$$f(x) = \begin{cases} x + 3, & x < 1 \\ 2x, & x \geq 1 \end{cases}$$

Q6. Define:

- (a) Jump Discontinuity
- (b) Removable Discontinuity

Advanced Level

Q7. Evaluate:

$$\lim_{x \rightarrow 0} \frac{e^x - 1}{x}$$

Q8. Find value of:

$$k$$

for continuity:

$$f(x) = \begin{cases} kx + 2, & x < 2 \\ 6, & x = 2 \\ x + k, & x > 2 \end{cases}$$

Q9. Determine type of discontinuity:

$$f(x) = \frac{1}{x - 5}$$